TAIL LEAKAGE: A SIMPLE CONFIDENCE SIGNAL FOR ADAPTIVE RETRIEVAL

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ABSTRACT

Adaptive retrieval requires a model to decide when its internal knowledge is insufficient. Token entropy is a natural training-free signal, but it mixes competition among plausible leading tokens with probability mass dispersed across the rest of the vocabulary. We introduce TAIL LEAKAGE, a confidence score that retains only the latter effect. At each generated position, it measures the negative log probability mass retained by the model's top-k tokens and sums this quantity over the answer. The score is available from one greedy decoding pass and requires neither sampling nor a learned router. We study TAIL LEAKAGE in SEEK, a dual-path open-domain question-answering framework that applies the score to sub-question retrieval and preserves both direct and decomposition candidates. Across four development-set traces with Llama-3-8B-Instruct, TAIL LEAKAGE reaches 0.749 macro AUROC for detecting wrong direct answers, compared with 0.726 for maximum token entropy. In the tested end-to-end configuration, replacing entropy with TAIL LEAKAGE raises macro EM/F1 from $40.52/47.61$ to $41.88/48.90$. Decomposition-path queries also increase answer-containing retrieval recall on three multi-hop datasets, while leaving single-hop PopQA nearly unchanged. These results support TAIL LEAKAGE as a useful ranking and gating signal, but confidently wrong answers remain a fundamental limitation.

1 INTRODUCTION

Retrieval-augmented generation (RAG) gives a language model access to information beyond its parameters by conditioning generation on external passages (Gao et al., 2023). A fixed retrieval policy is nevertheless poorly matched to heterogeneous questions. Retrieval can add latency and distracting evidence when the model already knows a simple fact, yet a single query is often insufficient for a multi-hop question whose decisive entity becomes known only after an intermediate step (Trivedi et al., 2023; Ren et al., 2023). Adaptive RAG therefore needs to decide not only *what* to retrieve, but also *when* the current reasoning state warrants retrieval.

One appealing solution is to use the language model's own token distribution. Uncertainty-triggered methods can act during generation without a separately trained routing model (Jiang et al., 2023; Su et al., 2024; Wang et al., 2025). In practice, however, uncertainty is not synonymous with factual error. A model can distribute probability among several harmless aliases or formatting variants while remaining factually correct; it can also assign almost all probability to a fluent hallucination. Standard entropy combines these cases because it records both competition among high-probability candidates and dispersion into the large low-probability tail.

This paper asks whether a more selective statistic of the token distribution can provide a better confidence signal for adaptive retrieval. We introduce TAIL LEAKAGE, which measures how much probability escapes the model's own small set of leading candidates. If m_t is the probability mass of the top-k tokens at generation step t, the answer-level score is

$$\mathrm{TL}(a) = -\sum_{t \in a} \log m_t.$$

The score is near zero when a small candidate set retains almost all probability and increases monotonically as mass moves into the tail. Unlike full entropy, it discards how probability is divided inside

the leading set and how the remaining mass is arranged within the tail. It is computed from the logits already produced during greedy decoding, without extra generations, labels, or router training.

We evaluate this signal inside SEEK, short for Self-certainty Evaluated Evidence-gated Knowledge-retrieval. This dual-path framework for open-domain question answering first produces a direct answer and also decomposes the original question into ordered, executable sub-questions. Each sub-question is tentatively answered without retrieval; TAIL LEAKAGE then decides whether to retrieve and re-answer that step. The decomposed steps are synthesized into a second candidate, and a transparent late rule selects between the direct and decomposition candidates. This design makes decomposition reversible: it can expose bridge entities for retrieval without discarding the original-question answer.

Our evaluation uses four development-set traces spanning three multi-hop datasets—HotpotQA, 2Wiki, and MuSiQue—and the single-hop PopQA benchmark. Among six representative confidence families, TAIL LEAKAGE gives the highest macro AUROC for detecting wrong direct answers (0.749, versus 0.726 for maximum token entropy). In the tested end-to-end configuration, the TAIL LEAKAGE system obtains $41.88/48.90$ macro EM/F1, compared with $40.52/47.61$ when maximum token entropy is used. The accuracy-first retrieval gate is effectively tied with always retrieving while skipping 6.16% of decomposed steps; more aggressive thresholds show a gradual accuracy–retrieval trade-off. The results do not imply that the score is a correctness test: confidently wrong answers remain, and final routing contributes only a small net gain.

Our contributions are:

- We derive TAIL LEAKAGE, a simple, monotonic, training-free statistic that isolates probability mass leaving the leading token set and accumulates it across an answer.
- We instantiate the score as a sub-question retrieval gate in SEEK, a reversible dual-path framework for decomposition-based RAG.
- We compare representative confidence signals, retrieval behavior, and end-to-end QA across four datasets, while identifying the boundary created by confident errors and imperfect path routing.

2 RELATED WORK

Adaptive retrieval. Adaptive-RAG routes questions to retrieval strategies using a learned complexity classifier (Jeong et al., 2024), while SKR estimates whether a model already knows an answer before retrieval (Wang et al., 2023). FLARE instead drafts the next sentence and retrieves when low-confidence tokens appear (Jiang et al., 2023); DRAGIN combines a generation-time trigger with attention-based query formulation (Su et al., 2024). TARG studies a lightweight question-level decision from prefix logits, including entropy and top-logit margins (Wang et al., 2025). These methods make uncertainty useful operationally, but most apply it either before reasoning or at the free-form token/sentence level. SEEK applies a training-free signal at explicit sub-question boundaries.

Learned policies provide finer control. Self-RAG trains reflection tokens that control retrieval and evaluate evidence (Asai et al., 2024). DeepRAG explores retrieve-or-answer decisions at individual sub-queries and calibrates the policy from preference data (Guan et al., 2026). Such policies can jointly learn reasoning and retrieval behavior, whereas our objective is complementary: a score that can be inspected and attached to an existing greedy generation without policy training.

Decomposition and multi-step retrieval. IRCoT interleaves chain-of-thought sentences with retrieval (Trivedi et al., 2023). Question-decomposition methods instead expose explicit intermediate operations or sub-questions (Min et al., 2019; Wolfson et al., 2020). Decomposition can create sharper retrieval queries, but a wrong bridge answer can corrupt later queries and short sub-questions can omit constraints from the original question. SEEK therefore retains a direct candidate and keeps original-question evidence available to the decomposition path. We treat these components as an evaluation framework for the confidence signal rather than attributing all system gains to decomposition alone.

Confidence and uncertainty in language models. Sequence probability, token entropy, verbalized self-evaluation such as P(True), and disagreement across sampled answers capture different aspects of model confidence (Kadavath et al., 2022; Kuhn et al., 2023; Farquhar et al., 2024). They are useful ranking signals, but a model may remain confidently wrong. TAIL LEAKAGE is closest to logit-based uncertainty methods, yet differs in what it retains: it ignores the exact probability arrangement within the head and tail and records only the mass crossing their boundary. This choice follows from an entropy decomposition in Section 3 and is tested against representative alternatives in Section 6.2.

3 TAIL LEAKAGE

Let $p_t(v)$ be the next-token distribution at position t during greedy generation, and let $V_k(p_t)$ contain its k most probable tokens. We define the retained mass

$$m_t^{(k)} = \sum_{v \in V_k(p_t)} p_t(v), \qquad \ell_t^{(k)} = 1 - m_t^{(k)}, \tag{1}$$

where $\ell_t^{(k)}$ is the mass that leaks outside the leading set. We use $k = 5$ throughout the main experiments and write m_t for $m_t^{(5)}$.

3.1 WHY ISOLATE LEAKED MASS?

Token entropy measures the total spread of p_t, but this spread has several sources. Let $\hat{p}_t^{\text{in}}$ and $\hat{p}_t^{\text{out}}$ denote the distributions obtained by renormalizing the tokens inside and outside $V_5(p_t)$, respectively. The entropy chain rule gives

$$\begin{aligned} H(p_t) = & -m_t \log m_t - (1 - m_t) \log(1 - m_t) \\ & + m_t H(\hat{p}_t^{\text{in}}) + (1 - m_t) H(\hat{p}_t^{\text{out}}). \end{aligned} \tag{2}$$

The first term records how much mass crosses the top-five boundary. The other terms record competition among the leading candidates and the arrangement of probability within the tail. For short-answer QA, competition among leading tokens can reflect casing, aliases, or tokenization rather than factual uncertainty. Conversely, the internal shape of a very large tail may change without materially changing the model's leading hypothesis. Equation 2 motivates testing the boundary-crossing mass separately instead of compressing all three effects into entropy.

3.2 TOKEN- AND ANSWER-LEVEL SCORE

We convert retained mass into a monotonic risk score:

$$\text{TL}_t = -\log m_t, \qquad \text{TL}(a) = \sum_{t \in \mathcal{T}(a)} \text{TL}_t = -\log \prod_{t \in \mathcal{T}(a)} m_t, \tag{3}$$

where $\mathcal{T}(a)$ contains the generated answer tokens and excludes EOS. The negative logarithm has two useful properties. First, it increases strictly as mass leaves the top-five set; unlike binary entropy, it does not decrease once leaked mass exceeds 0.5. Second, it is additive across positions, allowing small amounts of leakage distributed over a multi-token answer to accumulate.

TAIL LEAKAGE requires only the top-k probabilities already available during a greedy pass. It is therefore cheaper than sample-disagreement or semantic entropy estimates that require multiple generations. It is a ranking score, not a calibrated probability that an answer is wrong. In particular, if a model concentrates almost all probability on a false continuation, Equation 3 provides no warning.

4 ADAPTIVE RETRIEVAL WITH SEEK

We use SEEK to test whether TAIL LEAKAGE is useful beyond offline error detection. Given a question q, a language model $\mathcal{M}$, and a passage corpus $\mathcal{P}$, SEEK preserves two answer candidates until late routing:



Figure 1: The two candidate paths in SEEK. Decomposition is retained as a retrieval strategy rather than an irreversible early decision.

Executable decomposition. A LoRA-adapted decomposer maps q to ordered, knowledge-graph-style sub-questions. A step has the form `entity >> relation`; a placeholder such as `#1` refers to the answer of an earlier step. For example, `Inception >> director` followed by `#1 >> place of birth` becomes a directly executable second query after the first answer is substituted. One-step reformulations are allowed, which prevents the representation from forcing a multi-hop chain onto single-hop questions.

Step-level retrieval gate. For each resolved sub-question s_i, the model first generates a tentative answer $\tilde{a}_i$ without retrieved evidence and computes $\mathrm{TL}(\tilde{a}_i)$. Retrieval is invoked when

$$\pi(s_i) = \mathbb{I}[\mathrm{TL}(\tilde{a}_i) \geq \tau]\,. \tag{4}$$

If the gate fires, BM25 returns passages and the model re-answers s_i; otherwise $\tilde{a}_i$ is retained. Thus the signal operates at a natural reasoning boundary rather than at an arbitrary sentence or token position.

Context preservation and synthesis. Short entity-centered sub-questions are useful retrieval queries but can omit comparison targets or disambiguating constraints. When a step retrieves, SEEK therefore deduplicates the union of passages retrieved from s_i and from the original question q. After all placeholders are resolved, a base-model synthesis pass reads q and the ordered (s_i, a_i) pairs to produce a^{dec}. The direct candidate a^{dir} answers q from original-question evidence.

Late routing. A deterministic verifier defaults to the decomposition candidate and switches to a^{dir} only when direct-answer TAIL LEAKAGE is low and the decomposition path shows instability, such as high post-synthesis leakage, malformed placeholders, or an incompatible answer type. The verifier does not regenerate an answer or train a policy. We include it to preserve complementary candidates, but do not treat it as the principal source of accuracy: Section 6.4 shows that its net effect is small. Exact rules and diagnostic counts appear in Appendix A.

5 EXPERIMENTAL SETUP

Datasets. We evaluate on the first $1{,}000$ development questions, without shuffling, from each of HotpotQA (Yang et al., 2018), 2Wiki (Ho et al., 2020), MuSiQue (Trivedi et al., 2022), and PopQA (Mallen et al., 2023). The first three benchmarks contain multi-hop questions; PopQA consists of single-hop, entity-centric factual questions. The chosen MuSiQue slice contains only two-hop questions because the development set is ordered by hop count. We therefore report these results as a controlled development-set study, not as full-benchmark estimates.

Retrieval and generation. The corpus is an Elasticsearch BM25 index of approximately 21 million non-overlapping 100-word English Wikipedia passages following the DPR corpus construction (Karpukhin et al., 2020). Retrieval returns eight passages. Original questions match paragraph text; short decomposition queries match title and paragraph text. Each retrieved decomposition step also receives deduplicated evidence from the original question. This path-aware configuration is fixed across datasets. Because original and decomposed queries use different title fields, the retrieval comparison in Section 6.3 measures the complete decomposition-path query policy rather than decomposition wording in isolation.

All components use Llama-3-8B-Instruct with greedy decoding. The decomposer is trained once on $12{,}000$ examples: $4{,}000$ each from MuSiQue gold sub-questions, 2Wiki evidence-triple conversions, and PopQA one-step triples. Evaluation IDs are excluded from the PopQA training portion. Training

Table 1: EM/F1 compared with reported Llama-3-8B results.

Method	HotpotQA	2Wiki	MuSiQue	PopQA	Macro
CoT†	27.20/37.75	28.20/34.85	13.70/22.97	21.20/25.33	22.58/30.23
CoT-Retrieve†	34.90/46.85	35.80/43.41	19.10/28.18	32.80/45.87	30.65/41.08
FLARE†	23.80/32.88	30.30/37.45	14.50/23.57	28.80/40.61	24.35/33.63
DRAGIN†	27.60/38.05	29.10/35.68	11.80/19.97	22.60/28.53	22.78/30.56
DeepRAG†	**40.70/51.54**	48.10/53.25	22.50/30.40	42.50/47.80	38.45/45.75
SEEK	37.80/49.03	**49.30/56.22**	**26.50/34.63**	**53.90/55.73**	**41.88/48.90**

† Values reported by Guan et al. (2026); not rerun in our harness.

uses LoRA (Hu et al., 2022) with rank 16, scale 32, dropout 0.05, and all linear modules for three epochs. We use bf16, a 1,024-token context, effective batch size 16, learning rate 10^{-4}, cosine decay, and 10% warmup in LLaMA-Factory (Zheng et al., 2024). The adapter is disabled during final synthesis.

Operating point and metrics. The main retrieval gate uses $k = 5$ and the accuracy-first threshold $\tau = 0.002305$, which retrieves 93.84% of decomposed steps. This point was selected from a pooled threshold sweep on the same development traces. Accordingly, the threshold sweep is descriptive and should not be interpreted as held-out hyperparameter selection; disjoint calibration is a required confirmatory experiment. We report normalized exact match (EM), token-level F1, pooled step retrieval rate, answer-containing Hit@5, and AUROC for predicting whether a direct answer is wrong. Generation is deterministic and we do not report seed variance.

Comparisons. For confidence analysis, we compare TAIL LEAKAGE with maximum token entropy, sequence negative log-likelihood, sampled-answer disagreement, the TARG top-logit margin (Wang et al., 2025), and P(True) self-evaluation (Kadavath et al., 2022). For QA, we include Llama-3-8B results reported by DeepRAG (Guan et al., 2026) for CoT, CoT-Retrieve, FLARE, DRAGIN, and DeepRAG. Those values were not rerun in our harness, so implementation, prompting, and normalization differences prevent a strict controlled ranking. All SEEK ablations use saved traces from our own pipeline.

6 RESULTS

6.1 END-TO-END QUESTION ANSWERING

Table 1 compares SEEK with published Llama-3-8B results. SEEK has the highest reported average in this set and the strongest values on 2Wiki, MuSiQue, and PopQA, while remaining below DeepRAG on HotpotQA. The comparison is encouraging but not controlled: the baseline rows come from the DeepRAG paper and may differ in prompts, implementation, and scoring. We therefore do not claim a new state of the art. On PopQA, the decomposition candidate is a one-step query reformulation, not a multi-hop reasoning chain.

6.2 DOES TAIL LEAKAGE IDENTIFY ANSWER RISK?

Table 2 reports AUROC for predicting a wrong direct answer; larger values indicate better ranking. TAIL LEAKAGE has the strongest macro AUROC among the tested representatives (0.749), improving over maximum token entropy and sequence NLL (both 0.726). Its advantage is not uniform across every dataset: sequence NLL ties it on PopQA, and P(True) ties it on MuSiQue. The correct conclusion is therefore limited to these traces and the tested signal families.

The end-to-end comparison points in the same direction. Using maximum token entropy gives $40.52/47.61$ macro EM/F1, while the TAIL LEAKAGE configuration gives $41.88/48.90$. The gain is again non-uniform: maximum entropy is 0.70 EM higher on 2Wiki, while TAIL LEAKAGE is higher

Table 2: AUROC for detecting wrong direct answers.

Signal	HotpotQA	2Wiki	MuSiQue	PopQA	Macro
TAIL LEAKAGE	**.757**	**.717**	**.726**	**.798**	**.749**
Max token entropy	.743	.671	.711	.780	.726
Sequence NLL	.733	.667	.707	**.798**	.726
Sample disagreement	.678	.659	.680	.674	.672
TARG margin	.669	.589	.618	.682	.639
P(True)	.655	.567	**.726**	.579	.632

Table 3: Answer-containing Hit@5 for original and decomposition-path queries.

Dataset	N	Original	Decomposition path	Difference
2Wiki	917	32.61	**60.63**	+28.03
HotpotQA	941	47.40	**52.92**	+5.53
MuSiQue	1,000	19.00	**38.80**	+19.80
PopQA	1,000	**71.20**	71.10	-0.10

on the other three datasets. An entropy-component analysis in Appendix C provides mechanism-level evidence: the head-versus-tail mass component reaches 0.742 macro AUROC, whereas the arrangement within the tail is near chance at 0.502.

6.3 RETRIEVAL BEHAVIOR

Table 3 measures whether any top-five passage contains a gold answer or alias. The decomposition-path query policy substantially improves recall on the three multi-hop datasets and leaves PopQA almost unchanged. The largest gains occur on 2Wiki and MuSiQue, where bridge entities make the later query more specific. As noted in Section 5, the two paths also differ in title matching, so these values should not be attributed to decomposition wording alone.

The gate controls an accuracy–retrieval trade-off rather than eliminating most retrieval calls. At the accuracy-first point, it is tied with always retrieve to two decimal places while retrieving 93.84% of steps. Raising the threshold to 0.008461 lowers retrieval to 87.82% at a cost of 0.26 macro EM and 0.24 macro F1. More aggressive skipping causes larger degradation (Table 4). These are fixed-chain replays: stored step branches are reselected and the chain is resynthesized, while later resolved sub-questions remain fixed.

6.4 WHAT DOES LATE ROUTING ADD?

The two candidates are complementary, but decomposition supplies more unique correct answers on every dataset. Across the four $1{,}000$-question traces, the direct path alone is correct in 173 cases, whereas only the decomposition path is correct in 874 cases. Relative to always choosing decomposition, the verifier recovers 52 direct-only cases but loses 41 decomposition-only cases, for a net $+11$ correct answers ($+0.28$ macro EM). The gain is concentrated on HotpotQA; routing changes EM by -0.4, -0.2, and -0.1 on 2Wiki, MuSiQue, and PopQA, respectively. We therefore view the verifier as a conservative fallback, not the source of the main performance difference. Full overlap and oracle counts appear in Appendix D.

7 LIMITATIONS

The present evidence is diagnostic rather than a complete confirmatory evaluation. First, the operating threshold was selected on the same development traces used for reporting; a disjoint calibration/evaluation split is needed before drawing general conclusions about the selected point. Second, all results use one backbone, greedy decoding, and one run on the first $1{,}000$ development examples. The MuSiQue slice contains only two-hop questions and understates the difficulty of the complete development set. Third, external baselines are paper-reported rather than reproduced in the same harness. Fourth, original and decomposition-path retrieval differ in title matching, and

Table 4: Accuracy–retrieval trade-off in the fixed-chain gate replay.

Policy / threshold	Macro EM/F1	Step retrieval
Never retrieve	25.98/32.34	0.00%
$\tau = 0.002305$	**41.88/48.90**	93.84%
$\tau = 0.008461$	41.62/48.66	87.82%
$\tau = 0.044170$	40.60/47.72	75.51%
$\tau = 0.116500$	39.42/46.62	64.40%
Always retrieve	**41.88/48.90**	100.00%

original-question evidence union has not been isolated in a separate ablation. The retrieval-recall gain therefore belongs to the complete query policy, not uniquely to decomposition.

The threshold sweep is a fixed-chain replay rather than an independent live rerun at every point. It preserves a controlled comparison but cannot capture all downstream changes caused by an earlier retrieval decision. Finally, TAIL LEAKAGE measures the shape of a model distribution, not factual correctness. It cannot flag a false answer generated with extreme confidence, and routing cannot repair questions for which both candidates are wrong.

8 CONCLUSION

We introduced TAIL LEAKAGE, a training-free confidence statistic that isolates probability mass leaving a model's leading token set and accumulates it across an answer. In the evaluated Llama-3-8B traces, it ranks direct-answer errors better than representative entropy, likelihood, disagreement, margin, and self-evaluation signals, and it serves as a useful sub-question retrieval gate inside SEEK. The results also set a clear boundary: most decomposed steps still benefit from retrieval, late routing adds only a small correction, and confident hallucinations remain. Broader backbones, held-out calibration, and controlled retrieval ablations are the next tests required to establish how far the signal generalizes.

AI USE STATEMENT

Generative AI tools were used to help organize the manuscript, draft and edit prose, translate between Chinese and English, review code, format LaTeX, and assist with interpreting already-computed experimental results. The reported numbers were checked against saved experimental artifacts, and all AI-assisted text and code were reviewed by the authors. The authors take responsibility for the final content, including text and artifacts produced with AI assistance.

ETHICS STATEMENT

This work evaluates language models on existing public question-answering benchmarks and Wikipedia passages. It introduces no new human-subject data. Like other retrieval-augmented generation systems, the method can reproduce biases or factual errors present in the model and retrieval corpus; confidence scores should not be treated as guarantees of correctness.

REPRODUCIBILITY STATEMENT

Section 5 specifies the model, data slices, corpus, retrieval configuration, training setup, and metrics. Appendix A provides implementation details, Appendix B gives additional retrieval analyses, and Appendices C and D report confidence and routing diagnostics. The supplementary package is intended to include the scripts and saved traces needed to reproduce the reported tables.

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A Implementation Details

A.1 Training and inference configuration

Table 5: Configuration used for the reported SEEK runs.

Setting	Value
Backbone	Llama-3-8B-Instruct
Decomposer data	12,000 examples; 4,000 per source
LoRA	rank 16, scale 32, dropout 0.05
Target modules	all linear modules
Training	3 epochs, bf16, effective batch 16
Optimization	10^{-4}, cosine decay, 10% warmup
Maximum context	1,024 tokens
Decoding	greedy; at most 48 answer tokens
Retrieval	BM25, top 8, Wikipedia-21M
Tail set	top $k = 5$ tokens
Step gate	$\tau = 0.002305$
Direct verifier	$\tau_d = 0.0005$
Post-decomposition verifier	$\tau_p = 0.0158$
Type-aware margin	2.5

A.2 Prompts

The answer probe uses the same answer-only instruction for direct and sub-question generation:

> *Answer the question with the answer phrase only. No explanation, no preamble, no full sentence—just the answer.*

For a decomposed step, the original question is prepended when it differs from the resolved sub-question. Retrieved context, when available, appears between the original question and sub-question.

The decomposer instruction asks for an ordered list of atomic sub-questions, one per line, with the format `#<index>:`. Dependencies must use a `#N` placeholder. The final synthesizer receives the original question and all sub-question/answer pairs. It returns only a short answer span and contains explicit rules for comparison, yes/no, and year questions. The supplementary source package contains the exact strings and chat-template construction.

A.3 Final routing rule

The final rule is deterministic. It first handles exact candidate agreement, empty or malformed outputs, unresolved placeholders, and obvious yes/no or comparison-type mismatches. If these rules do not decide the route, it chooses direct only when

$$\mathrm{TL}(a^{\mathrm{dir}}) < \tau_d \quad \text{and} \quad \max_i \mathrm{TL}(a_i^{\mathrm{post}}) > \tau_p$$

or the decomposition is structurally broken; otherwise it chooses decomposition. A final type-aware check is applied only to undecided decomposition defaults. It scores answer-type compatibility, whether the final answer repeats an intermediate answer, whether a step answer appears as an entity fragment inside a later query, and whether a candidate simply repeats part of the question. It switches to direct only when the direct score exceeds the decomposition score by at least 2.5.

B Additional Retrieval Results

B.1 Gold and untuned decompositions on MuSiQue

The first 1,000 MuSiQue development questions are all two-hop, allowing comparison with the official decomposition. Table 6 shows that SEEK improves over the original query but remains below a gold chain. Replacing the bridge entity with the gold answer closes most of the gap, indicating that bridge-answer drift is more important than sub-question wording in this slice.

Table 6: Answer-containing Hit@5 on the MuSiQue slice.

Query condition	Hit@5
Original question	19.0
SEEK decomposition	38.8
Gold decomposition + gold bridge	**55.8**
Gold decomposition, unresolved `#1`	19.8
SEEK wording + gold bridge	54.1
Untuned base wording + gold bridge	52.9

B.2 Title matching

Original questions are long and constraint-rich, while decomposed queries are short and entity-centered. Table 7 compares title matching for both query types. These are live end-to-end reruns.

Table 7: EM/F1 under three BM25 title policies.

Dataset	Titles off	Titles on	Path-aware
PopQA	51.9/54.0	52.4/54.6	**53.4/55.7**
HotpotQA	37.3/48.7	37.1/48.4	**37.9/49.3**
MuSiQue	23.3/31.6	24.9/**34.5**	**26.2**/34.4
2Wiki	44.5/50.4	49.1/55.5	**49.6/56.3**

C Additional Confidence Results

C.1 Signal definitions

All scores are oriented so larger values indicate greater uncertainty.

- **Maximum token entropy**: the largest next-token entropy in the generated answer.
- **Sequence NLL**: the summed negative log-likelihood of the generated tokens.
- **TARG margin**: the transformed average gap between the two largest logits over the first eight answer tokens.
- **Sample disagreement**: the fraction of ten sampled answers that differ from the greedy answer.
- **P(True)**: one minus the model's estimated probability that its proposed answer is true.
- **TAIL LEAKAGE**: the sum of negative log top-five mass in Equation 3.

The first three and TAIL LEAKAGE use the same greedy pass. Sample disagreement and P(True) require additional generations.

C.2 ENTROPY COMPONENTS

Table 8 evaluates the three terms of Equation 2. The boundary-crossing component carries most of the useful signal; the arrangement within the tail is close to chance. These component values come from the saved diagnostic computation and should be recomputed from complete logits before being elevated to a primary claim.

Table 8: Macro AUROC of entropy components.

Component	Macro AUROC
Leading set versus tail	0.742
Competition within top five	0.712
Arrangement within tail	0.502
Full token entropy	0.726
Answer-level TAIL LEAKAGE	**0.749**

C.3 END-TO-END SIGNAL REPLACEMENT

Table 9: EM/F1 with entropy and Tail Leakage.

Signal	HotpotQA	2Wiki	MuSiQue	PopQA	Macro
Max entropy	35.50/46.11	**50.00/56.58**	24.70/33.06	51.90/54.71	40.52/47.61
TAIL LEAKAGE	**37.80/49.03**	49.30/56.22	**26.50/34.63**	**53.90/55.73**	**41.88/48.90**

C.4 SELECTIVE ANSWERING

Table 10 ranks direct answers by TAIL LEAKAGE and retains only the most confident fraction. Accuracy increases as coverage decreases, but the hard multi-hop datasets remain unreliable even at low coverage.

Table 10: Direct-path EM by confidence coverage.

Coverage	HotpotQA	2Wiki	MuSiQue	PopQA	Pooled
25%	57.6	34.0	22.4	79.2	52.8
50%	45.4	26.8	15.2	57.0	38.9
75%	38.0	22.1	12.5	46.3	30.2
100%	30.3	18.0	10.2	37.8	24.1

Table 11: Correctness overlap before final routing.

Dataset	Both correct	Direct only	Decomp. only	Neither
HotpotQA	225	78	135	562
2Wiki	144	36	353	467
MuSiQue	75	27	192	706
PopQA	346	32	194	428

Table 12: Verifier changes relative to always choosing decomposition.

Dataset	Direct-only recovered	Decomp.-only lost	Net EM
HotpotQA	39/78	21/135	+1.80
2Wiki	3/36	7/353	-0.40
MuSiQue	9/27	11/192	-0.20
PopQA	1/32	2/194	-0.10
Pooled / macro	52/173	41/874	+0.28

D Routing and Failure Analysis

D.1 Candidate overlap

D.2 Verifier effect

D.3 Decomposition-path failures

Each incorrect multi-hop decomposition candidate is assigned one primary cause. Evidence is counted as present when the required Wikipedia title appears in the top ten results from either the original or resolved sub-question queries.

Table 13: Primary causes of incorrect decomposition answers.

Cause	HotpotQA ($N = 640$)	2Wiki ($N = 503$)	MuSiQue ($N = 733$)
Missing required evidence	307 (48.0%)	311 (61.8%)	513 (70.0%)
Wrong step answer	131 (20.5%)	98 (19.5%)	73 (10.0%)
Wrong final synthesis	164 (25.6%)	92 (18.3%)	138 (18.8%)
Lost original constraint	38 (5.9%)	2 (0.4%)	9 (1.2%)

Missing evidence is the largest category on all three multi-hop datasets. Routing cannot repair many of these cases because the direct candidate is also wrong.

D.4 Confidently wrong cases

Both-wrong examples have higher scores on average, but the groups overlap. This is why Tail Leakage is used as a population-level ranking and gating signal rather than a per-question correctness test.

Table 14: Tail Leakage when both candidates agree in correctness.

Dataset	Both-correct mean	Both-wrong mean	AUROC	Confidently wrong
HotpotQA	0.01260	0.12927	0.777	15.5%
2Wiki	0.03414	0.20215	0.751	20.1%
MuSiQue	0.04945	0.25025	0.766	12.7%
PopQA	0.00974	0.10282	0.846	4.9%
Pooled	0.01877	0.17926	0.817	9.3%